WINDOWS SERVER ADMINISTRATION

Learning Outcome 3: Deploy DHCP services.

Key Reading 3.1.1: Description of DHCP Operations (Windows Server)

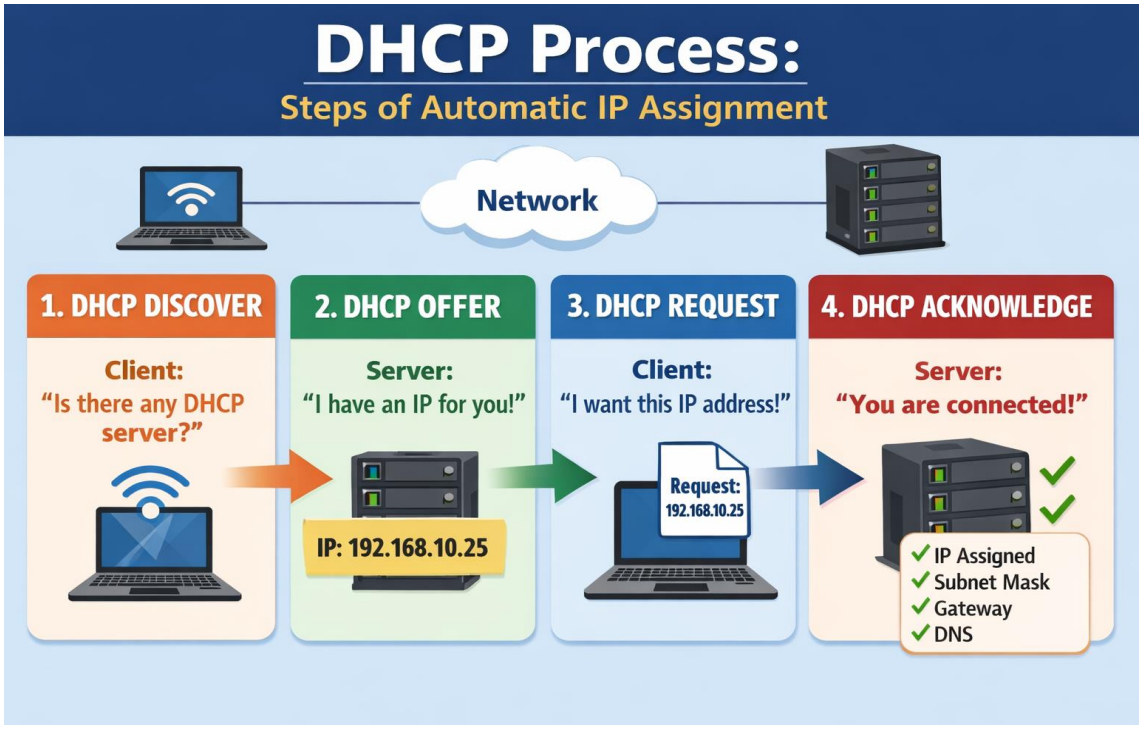
1. Introduction to Windows Server and DHCP

A **Windows Server** is a specialized operating system designed to manage network resources such as users, computers, files, printers, and network services.
One of the most important services provided by Windows Server is **DHCP (Dynamic Host Configuration Protocol)**.

A **DHCP Server** automatically gives network settings to computers and devices such as:

- IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time

Without DHCP, every device would need to be configured **manually**, which is slow and error-prone.



1.1 DHCP Discover (Client looks for a DHCP Server)

When a computer (client) starts for the first time on a network, it **does not have an IP address**.
To find a DHCP server, the client sends a **DHCP Discover** message.

How it works:

- The client broadcasts the message to **all devices** on the local network.
- It uses a **broadcast MAC address**:
  FF:FF:FF:FF:FF:FF
- This message asks:
  👉 *"Is there any DHCP server that can give me an IP address?"*

Example:

A student turns on a laptop in the computer lab.
The laptop has no IP address yet, so it sends a DHCP Discover message to the network.

Small Scenario:

Imagine entering a room and loudly asking:
"Is there anyone here who can help me?"
Everyone hears you. That is how DHCP Discover works.

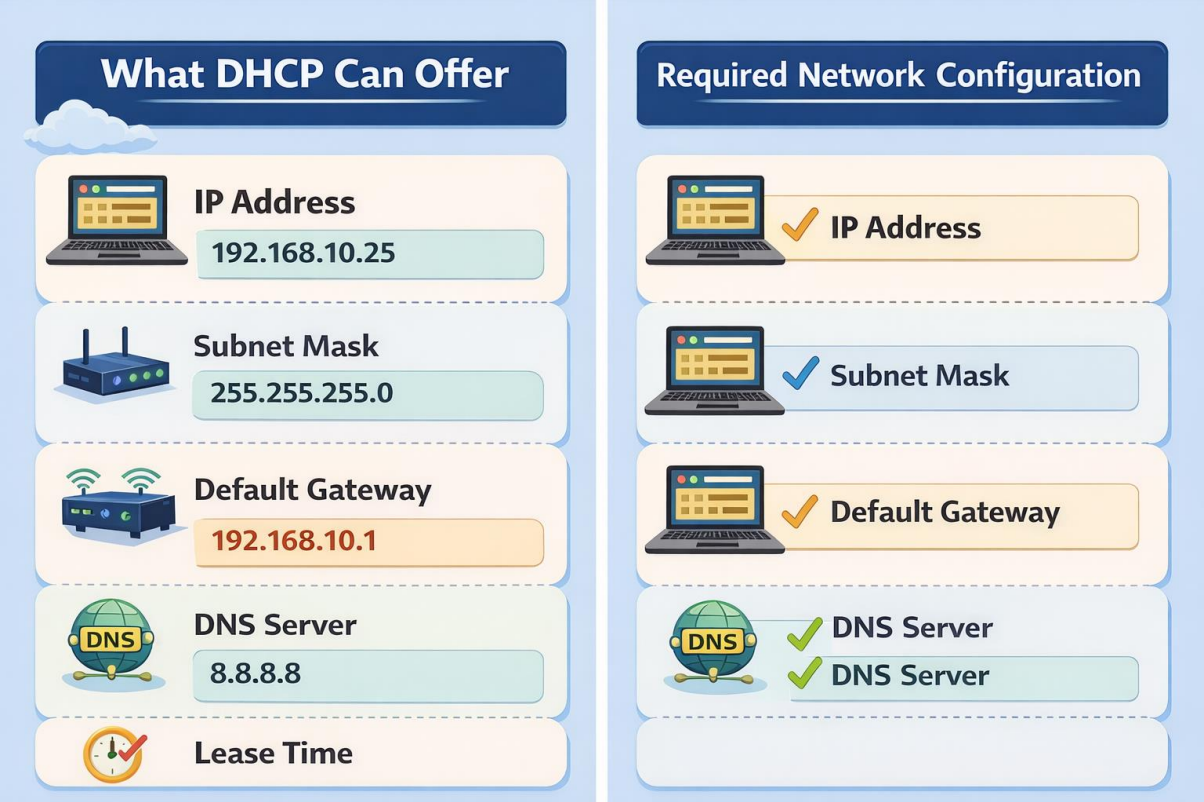
1.2 DHCP Offer (Server responds)

When a DHCP server receives the Discover message, it replies with a **DHCP Offer** message.

Important points:

- The DHCP Offer is also **broadcast** because the client still has no IP address.
- The message contains:
  - Proposed IP address
  - Subnet mask
  - Default gateway
  - DNS server address

- Lease time
- DHCP server IP address



All clients receive the offer, but **only the requesting client accepts it**.

Example:

The school DHCP server replies:

"I can give you IP address 192.168.10.25 for 24 hours."

Small Scenario:

It's like a teacher replying:
"Yes, I can help you. Here is what I can offer."

1 What DHCP Can Offer (Explanation)

1. IP Address

An **IP address** is a unique number given to a device on a network.
It helps the device to **identify itself** and communicate with other devices.

Example:
192.168.10.25
Just like a **student registration number**, no two devices should use the same IP at the same time.

2. Subnet Mask

The **subnet mask** tells the device which part of the IP address belongs to the network and which part belongs to the host.

Example:
255.255.255.0
This means all devices are in the **same local network**.

Simple idea:
It is like knowing **which class you belong to** in a school.

3. Default Gateway

The **default gateway** is usually the router.
It allows devices to communicate **outside the local network**, for example to access the Internet.

Example:
192.168.10.1

Simple idea:
It is the **main gate** of the school that allows students to go outside.

4. DNS Server

A **DNS server** converts website names into IP addresses.

Example:
8.8.8.8

Explanation:
When you type www.google.com, DNS finds the correct IP address for Google.

Simple idea:
DNS is like a **phonebook** for the Internet.

5. Lease Time

The **lease time** is the period a device is allowed to use an IP address.

Example:
24 hours

After the lease expires, the client must **renew** the IP address or get a new one.

Simple idea:
It is like **borrowing a book from the library for a limited time**.

2️⃣ Required Network Configuration (Explanation)

These are the settings a computer must have to **successfully connect to the network**.

✓ IP Address

The device must have a valid IP address to communicate on the network.

If there is **no IP address**, the device cannot talk to others.

✓ Subnet Mask

The subnet mask ensures the device knows **which network it belongs to**.

Wrong subnet mask = communication problems.

✓ Default Gateway

Without a default gateway:

- The device can talk inside the network
- ❌ But cannot access the Internet

✓ DNS Server

Without DNS:

- Internet may work
- ❌ But websites will not open using names
- ✓ **DHCP Server IP address:** is the IP address of the server that provides network configuration information to client devices.

🔁 Small Classroom Scenario

A student turns on a laptop in the computer lab.
The DHCP server automatically gives:

- IP address
- Subnet mask

- Default gateway
- DNS server

The **DHCP Server IP address** is the IP address of the server that provides network configuration information to client devices.

1.3 DHCP Request (Client chooses a server)

After receiving one or more DHCP Offers:

- The client selects **one DHCP server**.
- Then it broadcasts a **DHCP Request** message.

Why broadcast again?

- To inform **all DHCP servers** which offer was accepted.
- Other DHCP servers cancel their reserved IP addresses.

Special case: Multiple DHCP servers

If more than one DHCP server responds:

- The client chooses one.
- The chosen server's IP is written in **Option 54 (DHCP Server Identifier)**.

Example:

The client receives two offers:

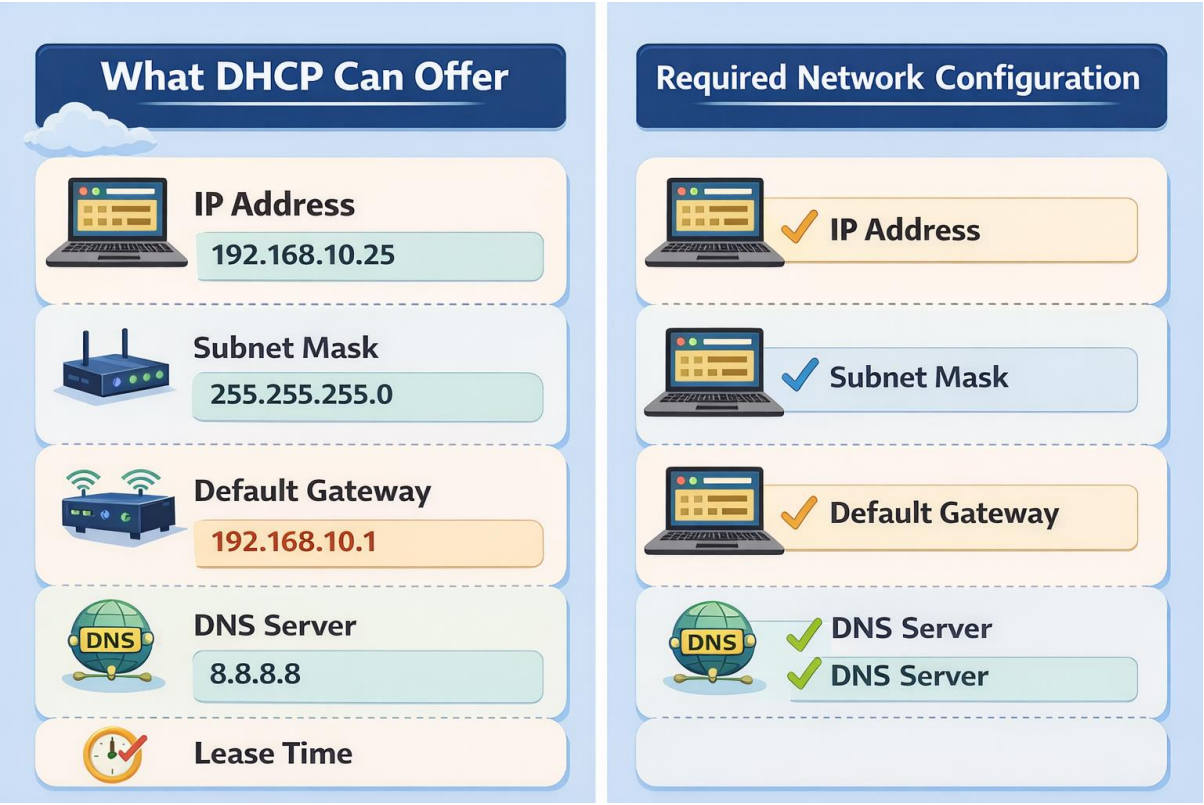
- Server A → 192.168.10.25
- Server B → 192.168.10.30

The client chooses Server A and sends a DHCP Request indicating Server A.

Small Scenario:

Like choosing one shop after receiving prices from many shops, then telling others:

“Thank you, I have chosen this one.”



1.4 DHCP Acknowledgement (Final confirmation)

The selected DHCP server checks the DHCP Request:

- If the server identifier matches its own, it responds with **DHCP ACK**.

DHCP ACK does the following:

- Confirms the IP address assignment
- Sends final network configuration details
- Allows the client to configure its network interface

After this step, the client can:

- Communicate on the network
- Access the Internet

Typical configuration sent:

- IP address
- Subnet mask
- Default gateway
- DNS server

- Lease time

Small Scenario:

This is like receiving final approval:
"Everything is confirmed. You can now use the network."

Key Reading 3.1.2: Performing DHCP Installation (Windows Server)

Why Install DHCP?

DHCP reduces workload for network administrators by **automating IP address assignment**.

Without DHCP:

- Every computer must be configured manually
- IP conflicts are common
- Network management becomes difficult

Prerequisites for DHCP Installation

Before installing DHCP, ensure:

- Windows Server (e.g., 2016, 2019)
- Static IPv4 address on the server
- Defined IP address range (scope)
- Administrator privileges

DHCP Installation Steps (Simplified)

Step 1: Open Server Manager

- Click **Start**
- Open **Server Manager**

Step 2: Add Roles and Features

- Click **Add roles and features**
- Click **Next**

Step 3: Choose Installation Type

- Select **Role-based or feature-based installation**
- Click **Next**

Step 4: Select Destination Server

- Choose the **local server**
- Click **Next**

Step 5: Select Server Role

- Check **DHCP Server**
- Click **Add Features**
- Click **Next**

Step 6: Features and DHCP Server

- Click **Next** until confirmation page

Step 7: Confirm Installation

- Click **Install**
- Wait for **Install succeeded**

👉 DHCP role is now installed.

Key Reading 3.2.1: Configuring DHCP (Microsoft Standard)

1. Enabling DHCP Client Service

- Open **services.msc**
- Locate **DHCP Client**
- Set Startup type to **Automatic**
- Click **Start**

2. DHCP Scope Configuration

What is a Scope?

A **scope** is a pool of IP addresses that DHCP can assign to clients.

Example Scope:

- Network: 192.168.100.0/24
- Range: 192.168.100.1 – 192.168.100.100

Steps to Create a Scope:

1. Open **DHCP Console**
2. Right-click server → **New Scope**
3. Give name and description
4. Define IP range
5. Exclude static IPs (servers, printers)
6. Set lease duration
7. Configure:
    - Default gateway
    - DNS server
8. Activate scope
9. Authorize DHCP server

Small Scenario:

Servers and printers have fixed seats.
Students (clients) get seats automatically.

3. DHCP Reservations

Reservations assign **fixed IPs based on MAC addresses**.

Example:

- Printer → always gets 192.168.100.10

4. Start DHCP Service

- Right-click DHCP Server
- Select **Start** or **Restart**

Key Reading 3.3.1: Testing DHCP Configuration

1. Verify DHCP Service

- Open **Server Manager**
- Tools → DHCP
- Confirm:

    - Service status: Started
    - Startup type: Automatic

PowerShell command:

Get-DhcpServerv4Lease -ScopeId <ScopeID>

2. Check Client IP Address

Command Line:

ipconfig /all

Look for:

- DHCP Enabled: Yes
- DHCP Server IP
- Lease obtained & expiry

Renew DHCP:

ipconfig /release

ipconfig /renew

3. GUI Method

- Open Network settings
- Check IPv4 details
- Confirm "Assigned by DHCP"

4. Troubleshooting Checklist

Verify:

- DHCP service is running
- Server is authorized

IP addresses are available

- No IP conflicts
- UDP ports 67 & 68 are free
- Event logs and DHCP logs are clean